

①

Transformer

- ① A 3300/250 V, 50 Hz, single phase transformer is built on a core having an effective cross-sectional area of 125 cm^2 and 70 turns on low voltage winding. Calculate: (a) the value of maximum flux density. (b) the no. of turns on high voltage winding.

Soln $\rightarrow E_1 = 3300 \text{ V}, E_2 = 250 \text{ V}, f = 50 \text{ Hz}, T_2 = 70$
 $A = 125 \text{ cm}^2 = 125 \times 10^{-4} \text{ m}^2$

$$E_2 = 4.44 \phi_m f T_2 = 4.44 B_m A f T_2$$

$$\therefore B_m = \frac{E_2}{4.44 A f T_2} = \frac{250}{4.44 \times (125 \times 10^{-4}) \times 50 \times 70}$$

$$= \boxed{1.289 \text{ T}}$$

$$\frac{E_1}{E_2} = \frac{T_1}{T_2}$$

$$\Rightarrow T_1 = \frac{E_1}{E_2} \times T_2 = \frac{3300 \times 70}{250} = \boxed{924}$$

- ② A transformer with 800 primary turns and 200 secondary turns is supplied from a 100 V a.c. supply. Calculate the secondary voltage and volts per turn.

Soln $\rightarrow T_1 = 800, T_2 = 200, V_1 = 100 \text{ V}$

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}, V_2 = V_1 \times \frac{T_2}{T_1} = 100 \times \frac{200}{800} = \boxed{25 \text{ V}}$$

$$\text{Volts/turn} = \frac{V_1}{T_1} = \frac{100}{800} = \boxed{0.125}$$

$$\text{or Volts/turn} = \frac{V_2}{T_2} = \frac{25}{200} = 0.125$$

3 A transformer with an output Voltage of 4200V is supplied at 230V. if the Secondary has 2000 turns, calculate the no. of primary turns.

Soln $\rightarrow V_2 = 4200V, V_1 = 230V, T_2 = 2000$

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$T_1 = \frac{V_1}{V_2} \times T_2 = \frac{230}{4200} \times 2000 = 109.52$$

In practice, it is not possible for a winding to have part of turn (i.e, turns cannot be fraction) Therefore, the no. of turns is the nearest whole number.

$$\therefore \boxed{T_1 = 110}$$

4 A 25 kVA transformer has a Voltage ratio of 3300/400 V. Calculate the primary and secondary currents

Soln $\rightarrow$ ~~25~~ $kVA = \frac{V_1 I_1}{1000}$

$$\Rightarrow I_1 = \frac{1000 kVA}{V_1} = \frac{1000 \times 25}{3300} = \boxed{7.58A}$$

$$kVA = \frac{V_2 I_2}{1000}$$

$$\Rightarrow I_2 = \frac{1000 \text{ kVA}}{V_2} = \frac{1000 \times 25}{400} = \boxed{62.5 \text{ A}}$$

- 5 A 125 kVA transformer having primary voltage of 2000 V at 50 Hz has 182 primary and 40 secondary turns. ~~Neglecting~~ Neglecting losses, calculate:
- full load primary and secondary currents
 - the no-load secondary induced emf
 - the maximum flux in the core

Solve →

$$(a) \text{ kVA} = \frac{V_1 I_1}{1000} = \frac{V_2 I_2}{1000}$$

$$I_1 = \frac{1000 \text{ kVA}}{V_1} = \frac{1000 \times 125}{2000} = \boxed{62.5}$$

$$I_2 T_2 = I_1 T_1$$

$$\Rightarrow I_2 = \frac{T_1}{T_2} \times I_1 = \frac{182}{40} \times 62.5 = \boxed{284.4 \text{ A}}$$

$$(b) \frac{E_2}{T_2} = \frac{E_1}{T_1}$$

$$E_2 = \frac{E_1 T_2}{T_1} = \frac{2000 \times 40}{182} = \boxed{439.6 \text{ V}}$$

$$(c) E = 4.44 \phi_m f T_1$$

$$\phi_m = \frac{E}{4.44 f T_1} = \frac{2000}{4.44 \times 50 \times 182} = \boxed{0.0495 \text{ wb}}$$

- 7 6 Find the active and reactive components of no-load current, and the no-load current of a 440/220V single phase transformer if the power input to the HV winding is 80 W. the low-voltage winding is kept open. the power factor of the no-load current is 0.3 lagging.

Soln $\rightarrow P_0 = V_1 I_0 \cos \phi_0$

Active Component of No-load current

$$I_W = I_0 \cos \phi = \frac{P_0}{V_1} = \frac{80}{440} = \boxed{0.182 \text{ A}}$$

No load current

$$I_0 = \frac{P_0}{V_1 \cos \phi_0} = \frac{80}{440 \times 0.3} = \boxed{0.606 \text{ A}}$$

Reactive component of No load current

$$\begin{aligned} I_M &= \sqrt{I_0^2 - I_W^2} \\ &= \sqrt{(0.606)^2 - (0.182)^2} = \boxed{0.578 \text{ A}} \end{aligned}$$

- 7 A 230/110V single phase transformer takes an input of 350 VA at no-load and at rated voltage. The core loss is 110 W. find (i) The iron loss component of no-load current. (ii) the magnetizing component of no-load current and (iii) No-load power factor

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Solve $\rightarrow V_1 I_0 = 350$

$$I_0 = \frac{350}{V_1} = \frac{350}{230} = 1.52 \text{ A}$$

$$\text{Core loss} = V_1 I_0 \cos \phi_0$$

$$110 = 350 \cos \phi_0$$

$$\cos \phi_0 = \frac{110}{350} = \boxed{0.314} \rightarrow \text{No load power factor.}$$

Core loss component of no-load current

$$I_w = I_0 \cos \phi_0 = \frac{\text{Core loss}}{V_1} = \frac{110}{230} = \boxed{0.478 \text{ A}}$$

Magnetizing component of no-load current

$$I_m = \sqrt{I_0^2 - I_w^2}$$

$$= \sqrt{(1.52)^2 - (0.478)^2}$$

$$= \boxed{1.44 \text{ A}}$$

Induction Motor

(6)

① A 12 pole, 3 phase alternator is coupled to an engine running at 500 r.p.m. It supplies an induction motor which has full load speed of 1440 rpm. find the slip and no. of poles of motor.

Solu → Engine running at 500 rpm
 $p = 12$

$$N_s = \frac{120f}{p} \Rightarrow f = \frac{p N_s}{120}$$

$$\Rightarrow f = \frac{12 \times 500}{120} = 50 \text{ Hz}$$

Therefore, supply frequency of induction motor is 50 Hz

the possible synchronous speed of Induction motor for different combination of poles → 2, 4, 6 ... are 3000, 1500, 750 rpm.

under normal operating condition an Induction motor operates at a speed slightly less than its synchronous speed. Therefore, the nearest speed to 1440 rpm is 1500 rpm.

$$N_s = 1500 \text{ rpm}$$

$$N = 1440 \text{ rpm}$$

$$\text{slip, } s = \frac{N_s - N}{N_s} = \frac{1500 - 1440}{1500}$$

$$s = \boxed{0.04 = 4\%}$$

$$p = \frac{120f}{N_s} = \frac{120 \times 50}{1500} = \boxed{4}$$

(7)

② The frequency of emf in rotor of a 4 pole induction motor is 50 Hz, and that of stator is 1.5 Hz. What is the slip, and at what speed is the motor running?

Solve → $f_2 = s f_1$
 $s = \frac{f_2}{f_1} = \frac{1.5}{50} = \boxed{0.03 = 3\%}$

$$N_s = \frac{120 f_1}{P} = \frac{120 \times 50}{4} = 1500 \text{ r.p.m.}$$

Speed of motor

$$N = N_s (1 - s) = 1500 (1 - 0.03) = \boxed{1455 \text{ r.p.m.}}$$

③ A 3 phase, 6 pole, 50 Hz induction motor has a slip of 1% at no load and 3% at full load. Determine

(a) Synchronous speed (b) No load speed (c) full load speed

(d) frequency of rotor current at ~~full load~~ standstill

(e) frequency of rotor current at full load.

Solution → (a) $N_s = \frac{120 f_1}{P} = \frac{120 \times 50}{6} = \boxed{1000 \text{ r.p.m.}}$

(b) $N_0 = (1 - s_0) N_s = (1 - 0.01) 1000 = \boxed{990 \text{ r.p.m.}}$

(c) $N_{fl} = (1 - s_{fl}) N_s = (1 - 0.03) 1000 = \boxed{970 \text{ r.p.m.}}$

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(d) frequency of rotor current at standstill
 $f_2 = s f_1 = 1 \times 50 = \boxed{50 \text{ Hz}}$

(e) frequency of rotor current at full load
 $f_2 = s_{fl} f_1 = 0.03 \times 50 = \boxed{1.5 \text{ Hz}}$

4 A 3-phase, 50 Hz Induction motor has 6 poles and operates with a slip of 5% at a certain load. Determine

- (i) the speed of rotor with respect to the stator
- (ii) the frequency of rotor current
- (iii) the speed of rotor magnetic field with respect to the rotor,
- (iv) the speed of rotor magnetic field with respect to the stator.

Soln $\rightarrow f_1 = 50 \text{ Hz}, P = 6$
 $s = 5\% = 0.05$

$$N_s = \frac{120 f_1}{P} = \frac{120 \times 50}{6} = 1000 \text{ r.p.m.}$$

(i) speed of rotor with respect to stator
 $N_1 = N_s (1 - s) = \frac{1000 (1 - 0.05)}{1} = \boxed{950 \text{ r.p.m.}}$

(ii) Frequency of rotor current
 $f_2 = s f_1 = 0.05 \times 50 = \boxed{2.5 \text{ Hz}}$

(iii) speed of rotor magnetic field w.r.t rotor
 $= \text{slip speed} = 1000 - 950 = \boxed{50 \text{ r.p.m.}}$

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(iv) speed of rotor magnetic field with respect to stator
 = synchronous speed
 = $N_s = \boxed{1000 \text{ r.p.m.}}$

(5) The Voltage applied to the stator of a 3-phase Induction motor is ~~480V~~ 4 pole motor has a frequency of emf induced in the rotor of 4 Hz. Calculate the slip and actual speed of the motor at running condition.

Solve $\rightarrow N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$

slip, $s = \frac{f_r}{f} = \frac{4}{50} = \boxed{0.08}$

actual speed, $N = N_s(1-s) = 1500(1-0.08)$
 $= \boxed{1380 \text{ rpm}}$

(6) A 3 phase, 4 pole, 50 Hz Induction motor runs at 1380 rpm. Determine the percentage slip of Induction motor.

Solve $\rightarrow N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$

$s = \frac{N_s - N_r}{N_s} = \frac{1500 - 1380}{1500} = 0.08$
 $= \boxed{8\%}$

(10)
7 A 3 phase, 8 pole induction motor is supplied by a 4 pole alternator which is driven at 1200 rpm. if the motor runs with a slip of 4%. what is its speed?

Soln $\rightarrow$ for the case of alternator
$$f = \frac{NP}{120} = \frac{1200 \times 4}{120} = 40 \text{ Hz}$$

for the case of induction motor

$$N_s = \frac{120f}{P} = \frac{120 \times 40}{8} = \boxed{600 \text{ r.p.m}}$$

$$\text{Now, slip } s = \frac{N_s - N_r}{N_s} = \frac{600 - N_r}{600}$$

$$0.04 = \frac{600 - N_r}{600}$$

$$N_r = \boxed{576 \text{ rpm}}$$